Test report of based on BSI AIS 20 / AIS 31

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1 Identification information

1.1 Identification information of input data

- Filename of input data : See Annex A.1.
- Name of the submitter of the input data :
- Brief explanation of the input data :

1.2 Identification of analysis environment

Table 1 Identification information of analysis environment

Analysis tool	Name	Another AIS 20/AIS 31 test tool
	Versioning information	3.0.2
	built as	64-bit application
	built by	Visual Studio 2019 version 16.11 (`_MSC_FULL_VER`: 192930157)
	linked libraries	Boost C++ 1.87.0
	with	OpenMP enabled
Analysis environment	Hostname	[redacted]
	CPU information	AMD Ryzen [redacted]
	Physical memory size	[redacted] MiB
	OS name	Microsoft Windows 11 Pro
	OS version	10.0.22631 N/A Build 22631
	System type	64-bit
	Username	[redacted]

1.3 Identification of analysis conditions

Table 2 Identification information of analysis conditions

Bits per sample	8
Byte to bit conversion	Least Significant bit (LSb) first

1.4 Identification of analysis method

Black Box Test Suite T_{irn} of BSI AIS 20 / AIS 31 [1] with corrections [2] is applied.

2 Executive summary

2.1 Test results based on BSI AIS 20 / AIS 31

Table 3 Test results

Tests	1-st trial		2-nd trial	
	Pass / Fail	Notes	Pass / Fail	Notes
Test T1 (monobit test)	Pass	see 3.1.1	N/A	—
Test T2 (poker test)	Pass	see 3.2.1	N/A	—
Test T3 (MultiMMC Prediction Estimate)	Pass	see 3.3.1	N/A	—
Test T4 (LZ78Y Prediction Estimate)	Pass	see 3.4.1	N/A	—
Overall test result	Pass			

3 Detailed information of tests for given input data

3.1 Test T1 (monobit test)

3.1.1 Supplemental information for traceability for the 1st trial

Table 4 Supplemental information for traceability (Test T1) for the 1-st trial

Symbol	Value
$t_{IRN(1)}$	9994

3.2 Test T2 (poker test)

3.2.1 Supplemental information for traceability for the 1st trial

Table 5 Supplemental information for traceability (Test T2) for the 1-st trial

Symbol	Value
$t_{IRN(2)}$	14.7392

3.3 The MultiMMC Prediction Estimate (Test T3)

3.3.1 Distribution of *correct* for the 1st trial

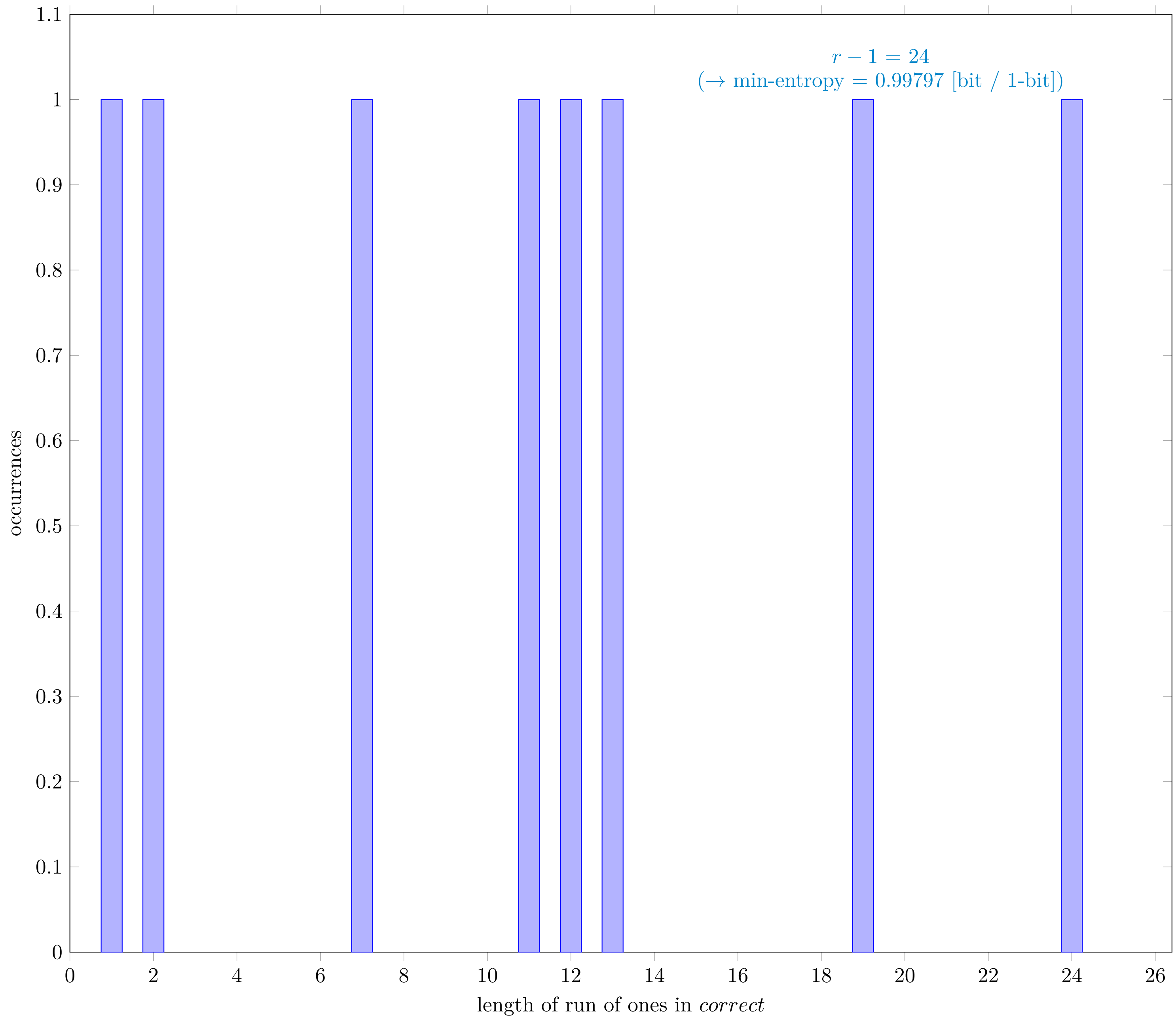

Fig. 1 Distribution of *correct*

3.3.2 Supplemental information for traceability for the 1st trial

Table 6 Supplemental information for traceability (NIST SP 800-90B Section 6.3.9)for the 1-st trial

Symbol	Value
N	999998
C	499415
P_{global}	0.499416
P'_{global}	0.500704
r	25
P_{local}	0.491868

3.4 The LZ78Y Prediction Estimate (Test T4)

3.4.1 Distribution of *correct* for the 1st trial

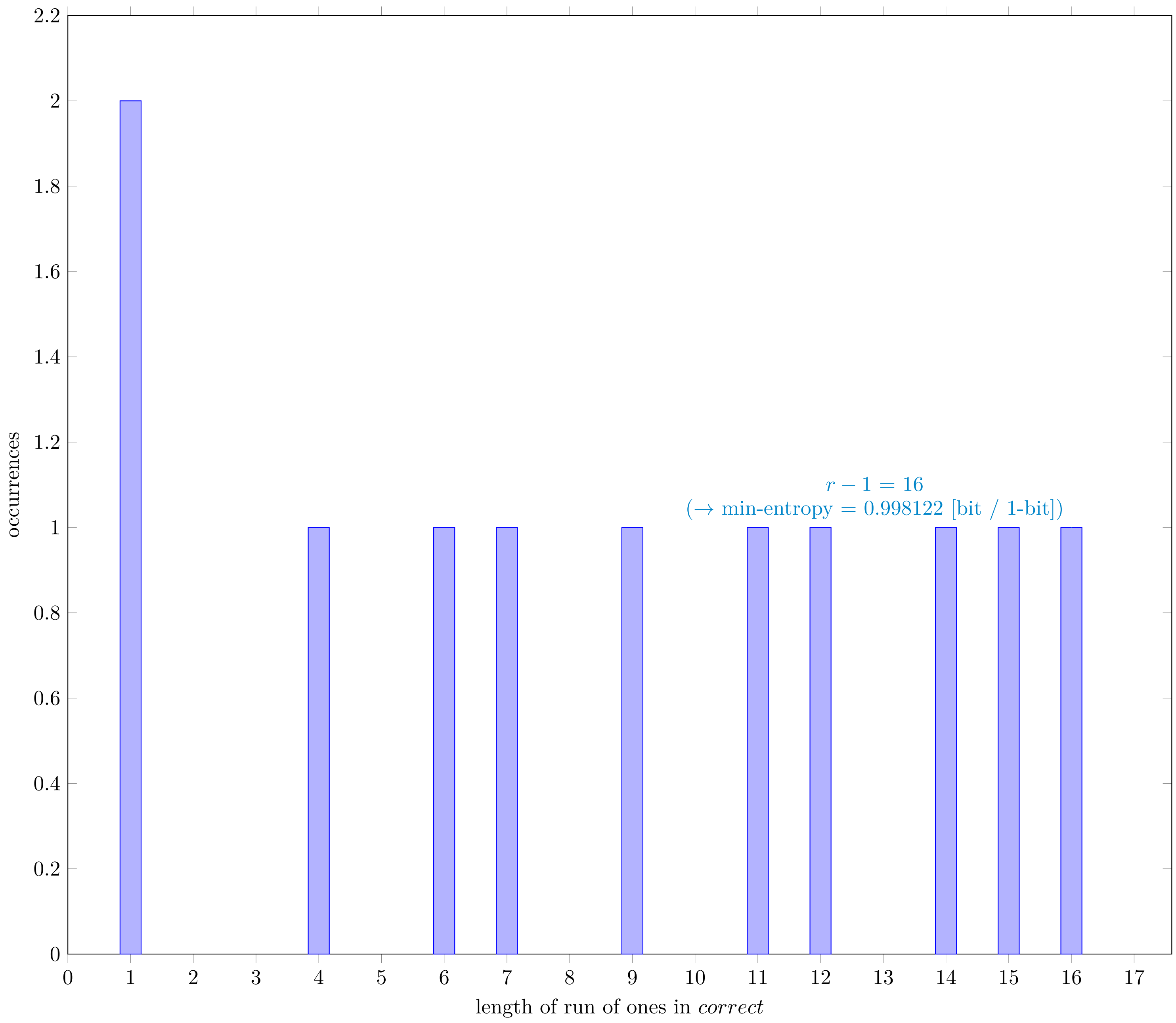

Fig. 2 Distribution of *correct*

3.4.2 Supplemental information for traceability for the 1st trial

Table 7 Supplemental information for traceability (NIST SP 800-90B Section 6.3.10)for the 1-st trial

Symbol	Value
N	999983
C	499355
P_{global}	0.499363
P'_{global}	0.500651
r	17
P_{local}	0.347082

A Identification information

A.1 Identification of input data

Table 8: Identification information of input data

No	Item	Value
1	URL of the input data for Tests T1 through T4	https://github.com/g-g-sakura/AnotherAIS31TestTool/tree/main/sample_input_data/v3_sample_input_data_for_T1-T4.bin
	SHA-256 hash value of the input data for Tests T1 through T4[hex]	a3b04689 c5512355 ba82b434 ea389f77 fd2883e2 b49d6c1b 479788b3 622dfdc1

end of the table

A References

[1] Matthias Peter and Werner Schindler. *A Proposal for Functionality Classes for Random Number Generators*, Version 3.0 (September 10, 2024), https://www.bsi.bund.de/SharedDocs/Downloads/EN/BSI/Certification/Interpretations/AIS_31_Functionality_classes_for_random_number_generators_e_2024.pdf?__blob=publicationFile&v=3

[2] G. Sakurai, *Proposed list of corrections for NIST SP 800-90B 6.3 Estimators*, Dec. 2022 https://github.com/g-g-sakura/AnotherEntropyEstimationTool/blob/main/documentation/ProposedListOfCorrections_SP800-90B.pdf